



Geological Survey Bulletin 1161-D

Geologic Reconnaissance of the Antelope-Ashwood Area, North-Central Oregon

PRIDAY AGATE DEPOSIT

The Priday agate deposit, located south of Pony Butte in sec. 35, T. 9 S., R. 15 E. (inset map, pl. 1), has yielded thousands of chalcedony-filled spherulites (thunder-eggs). The deposit is about 10 miles from Ashwood and 6 miles from U.S. Highway 97 on a county road that is only partly graveled. Most of the spherulites are taken from a layer that is 10 to 20 feet below the ground surface; the layer is exposed in an open pit by bulldozing and blasting, and the spherulites dug by hand.

The spherulites are chiefly in the lower few feet of a weakly welded rhyolitic ash flow (fig. 8) at the base of member F of the John Day Formation, but some are in pumice lapilli higher in the ash flow and in the upper few inches of the underlying stony ash-flow sheet (number E). The weakly welded ash flow (described on p. 10), which is 10 to 20 feet thick at the deposit, is composed of black perlitic angular lapilli of collapsed pumice in a matrix of shards and ash. Locally the basal part of the ash flow is altered to clay and to less abundant opal. Chalcedony-filled spherulites are widely distributed at this horizon; formerly they were recovered in large numbers from a locality about 1 mile northeast of the present Priday deposit.



FIGURE 8.—Spherulite-bearing ash flow at the base of member F of the John Day Formation, exposed in the open pit of the Priday agate deposit. Chalcedony-filled spherulites (thunder-eggs) occur near the base of the ash flow. The excavation in shadow near the bottom of the photograph was dug in recovering the spherulites.

The Priday deposit is on a low mesa supported by the resistant ash-flow sheet of member E and is surrounded by the underlying tuffs of member D (inset map, pl. 1). The spherulite horizon is preserved in a northward trending graben in which the Priday ash flow has been downdropped about 100 feet. The fault bounding the west side of the graben is exposed about 100 feet (in 1958) west of the pit, where it trends N. 10° E. and dips 80° E. West of this fault the Priday ash flow with its enclosed spherulites has been uplifted and eroded.

The thunder-eggs from the Priday deposit and other localities in Oregon and Idaho have been described in detail and illustrated by Dake (1938), Ross (1941), Renton (1951) and Brown (1957), so that a brief description will suffice here. They are small spheroidal bodies, about 3 inches in average diameter, and have a cauliflowerlike surface crossed by low ridges. Most consist of an outer shell of pale-brown aphanitic rock and a core of white to bluish-gray chalcedony. The outer shell of each thunder-egg is composed chiefly of shards, fine ash, and collapsed pumice lapilli, all of which are altered to radially oriented sheaves of fibrous cristobalite and alkalic feldspar. The chalcedonic cores commonly contain concentric and planar bands and dendritic mineral growths and range in shape from round and highly irregular forms to geometrically regular pyritohedrons and cubes, each face of which is an inward-pointing pyramid (Ross, 1941, pl. 2, fig. d; Brown, 1957, pl. 3, fig. 3). These appear as squares and stars in section.

A reasonable explanation of the origin of the thunder-eggs has been advanced by Ross (1941, p. 732). He concluded that spherulites formed during cooling of the ash flow and were disrupted by the pressure of volatiles exsolved from the ash; the resultant cavities were later filled by chalcedony during alteration of the enclosing ash flow.

[<<< Previous](#)

[<<< Contents >>>](#)

[Next >>>](#)

/sec3.htm

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